

Seatwork 7.1	
Using Sorting Algorithms	
Course Code: CPE010	Program: Computer Engineering
Course Title: Data Structures and Algorithms	Date Performed: Sep 15, 2025
Section: CPE21S4	Date Submitted: Sep 15, 2025
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Answer the following questions:	
1. What is a sorting algorithm? Sorting algorithm rearranges the data increasingly or decreasingly, depending on how it is sorted or arrange it into particular order. Data is swapped until it is sorted in particular order, it can be sorted in so many ways by using bubble sort, selection sort, and or insertion sort and many more. 2. Where can sorting algorithms be used? It can be used when we want to arrange data into particular order. It could be from smallest to biggest, biggest to smallest, by color, or and by value. There are many ways on how we sort data like how we sort our things. 3. Explain the different types of sorting algorithms. For now, There are 3 different types of sorting algorithms: Bubble sort, Selection sort, and Insertion sort. Bubble sort checks the element from the first element and compares it into the next element where if the first element is bigger than the next element, it swaps until the higher element reaches the last position and repeats until they are sorted into a particular order. Selection sort checks the data and finds and selects the smallest element and swaps it into the first element. Insertion sort, it is most likely the same to bubble sort but it does not repeat. It is until it finds the element that it should be the next to the first element and swaps it. 4. Give sample programs in C++ that uses sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the programs work.	

Bubble Sort:

```
#include <iostream>
using namespace std;

void bubbleSort(int arr[], int n) {
    for (int i = 0; i < n-1; i++) {
        for (int j = 0; j < n-i-1; j++) {
            if (arr[j] > arr[j+1]) {
                swap(arr[j], arr[j+1]);
            }
        }
    }
}

void printArray(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

int main() {
    int arr[] = {4, 2, 7, 1};
    int n = sizeof(arr)/sizeof(arr[0]);
    bubbleSort(arr, n);
    cout << "Sorted array: \n";
    printArray(arr, n);
    return 0;
}
```

Selection Sort:

```
#include <iostream>
using namespace std;

void swap(int *xp, int *yp) {
    int temp = *xp;
    *xp = *yp;
    *yp = temp;
}

void selectionSort(int arr[], int n) {
    int i, j, min_idx;
    for (i = 0; i < n-1; i++) {
        min_idx = i;
        for (j = i+1; j < n; j++)
            if (arr[j] < arr[min_idx])
                min_idx = j;
        swap(&arr[min_idx], &arr[i]);
    }
}

void printArray(int arr[], int size) {
    for (int i=0; i < size; i++)
        cout << arr[i] << " ";
    cout << endl;
}

int main() {
    int arr[] = {64, 25, 12, 22, 11};
    int n = sizeof(arr)/sizeof(arr[0]);
    selectionSort(arr, n);
    cout << "Sorted array is: \n";
    printArray(arr, n);
    return 0;
}
```

Insertion Sort:

```
#include <iostream>
using namespace std;

void insertionSort(int arr[], int n) {
    int i, key, j;
    for (i = 1; i < n; i++) {
        key = arr[i];
        j = i - 1;
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j = j - 1;
        }
        arr[j + 1] = key;
    }
}

void printArray(int arr[], int n) {
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
}

int main() {
    int arr[] = {12, 11, 13, 5, 6};
    int n = sizeof(arr) / sizeof(arr[0]);
    insertionSort(arr, n);
    cout << "Sorted array is: ";
    printArray(arr, n);
    return 0;
}
```

Reference

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